DECEMBER

The Sun reaches its farthest point south of the celestial equator this month, on December 21–22. As a result, Northern Hemisphere nights are the longest of the year, while in the Southern Hemisphere, they are the shortest. Earth has now completed another annual circuit of the Sun, and the evening stars end the year as they began, with the tableau of Orion and Taurus returning to center stage.

SPECIAL EVENTS

PHASES OF THE MOON

	FULL MOON	NEW MOON
2020	December 30	December 14
2021	December 19	December 4
2022	December 8	December 23
2023	December 27	December 12
2024	December 15	December 1, 30
2025	December 4	December 20
2026	December 24	December 9
2027	December 13	December 27

PLANETS

2020: December 21 Jupiter and Saturn appear less than a quarter of a Moon-width apart in the low western dusk sky.

2021: December 29 Mercury and Venus appear about eight Moon-widths apart in the low western dusk sky.

2022: December 8 Mars is at opposition, magnitude –1.8. At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2022: December 21 Mercury is at greatest evening elongation, magnitude –0.3.

2023: December 4 Mercury is at greatest evening elongation, magnitude –0.2.

2024: December 7 Jupiter is at opposition, magnitude –2.8.

2024: December 25 Mercury is at greatest morning elongation, magnitude –0.1.

2025: December 7 Mercury is at greatest morning elongation, magnitude –0.2.

2028: December 31 Mercury is at greatest evening elongation, magnitude –0.3.

ECLIPSES

2020: December 14 A total eclipse of the Sun is visible from the south Pacific, Chile, Argentina, and the south Atlantic. A partial eclipse is visible from the Pacific, southern South America, and Antarctica.

2021: December 4 A total eclipse of the Sun is visible from Antarctica. A partial eclipse is visible from Antarctica, South Africa, and the southern Atlantic.

2028: December 31 A total eclipse of the Moon is visible across the Arctic, eastern Europe, Asia, and Australasia.

NORTHERN LATITUDES

THE STARS

Overhead lies Perseus, containing the famous variable star Algol (see p.276). From Perseus, the Milky Way leads northwestward to Cassiopeia and Cygnus, which is out of sight for those at around 20°N or closer to the equator. In the other direction, the Milky Way extends southeastward via Auriga and past Taurus to Gemini and the northern arm of Orion. The Square of Pegasus is in the west, while the Winter Triangle of Betelgeuse (see p.256) in Orion, Procyon (see p.284) in Canis Minor, and Sirius (see p.252) in Canis Major dominates the southeast. By comparison with the richness of this southeastern part of the sky, the southwest seems dull and empty, as it is occupied by the faint constellations Aries, Pisces, and Cetus. As the year ends, Sirius lies due south around midnight.

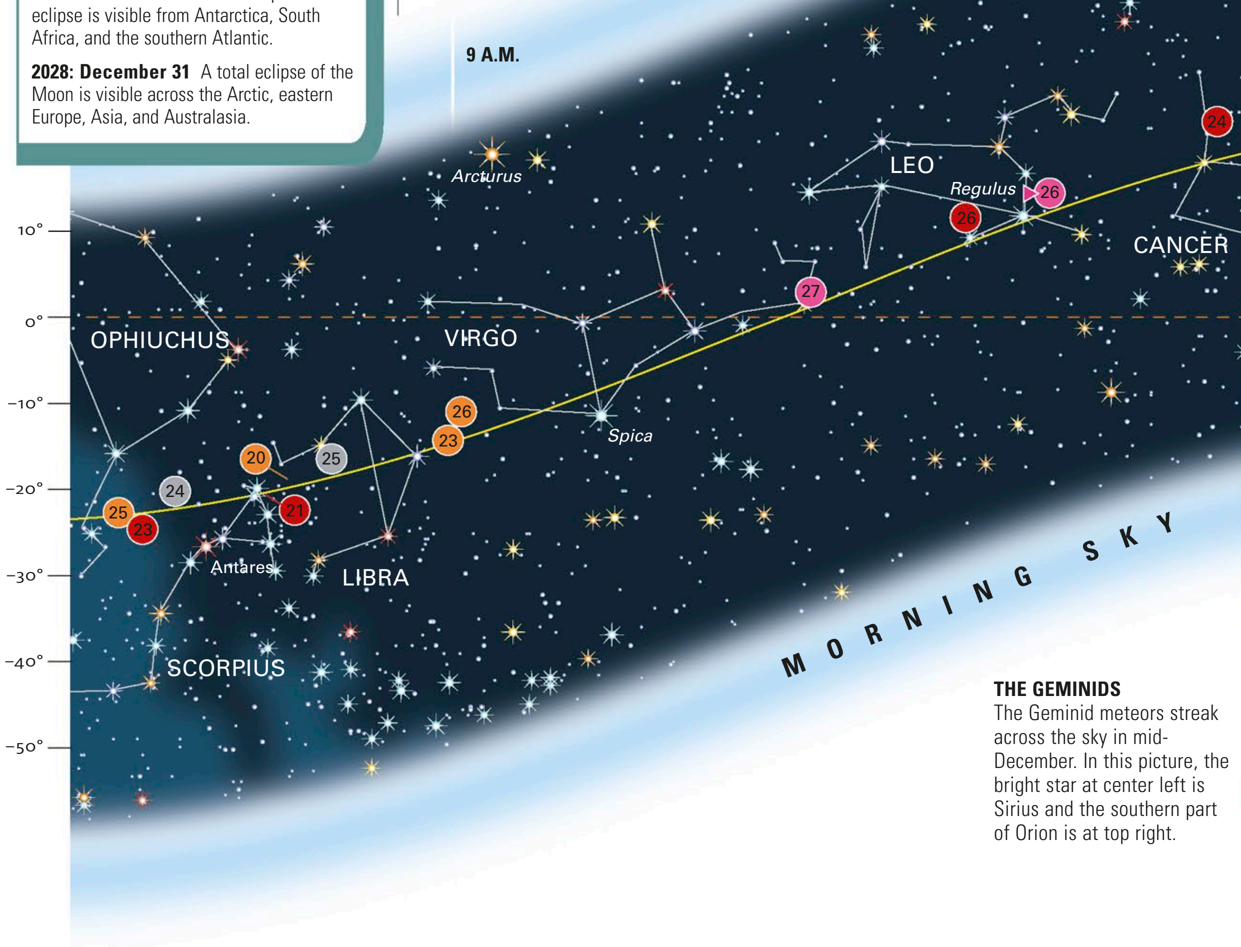
DEEP-SKY OBJECTS

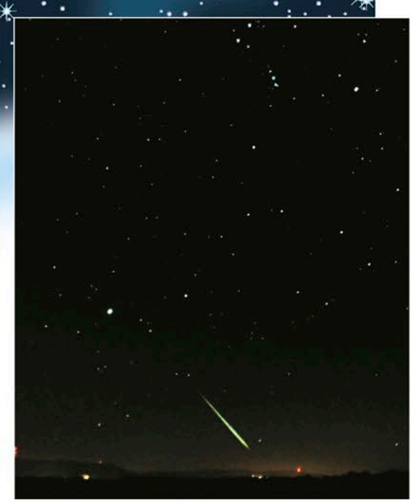
Large, bright clusters of stars abound in the December evening sky. In central Perseus, a few dozen stars cluster around the constellation's brightest member, Alpha (α) Persei or Mirfak. They form a group known as the Alpha Persei Cluster, which covers several diameters of a full moon and is a fine sight through binoculars.

In Taurus lies probably the finest open cluster in the entire sky, the Pleiades or M45 (see p.291). At least six members are visible to normal eyesight, but binoculars bring dozens more into view. Taurus contains an even larger cluster, the Hyades (see p.290), a V-shaped grouping which outlines the Bull's face. In addition to these groupings, the Double Cluster in Perseus, NGC 869 and NGC 884, already encountered in November, remains well placed.

METEOR SHOWER

The year's second-best meteor shower, the Geminids, reaches a peak around December 13–14, when up to one meteor per minute can be seen radiating from a point near Castor in Gemini. Lesser activity is seen for a few days before the peak, but numbers fall off rapidly afterward.





THE GEMINIDS The Geminid meteors streak across the sky in mid-December. In this picture, the bright star at center left is Sirius and the southern part of Orion is at top right.